

Hardware, Software & Firmware

Lesson 17 — The Layered Computing Stack

JKSSB Computer Awareness

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What Is Hardware?

- ▶ **Hardware** means the **physical, tangible** parts of a computer — the parts you can see and touch.
- ▶ Examples: the CPU, motherboard, RAM, hard disk, keyboard, mouse, monitor, and printer.
- ▶ Hardware by itself does nothing useful; it needs instructions to operate.
- ▶ It is the part of the system that physically **executes** work.

Definition

Hardware is the physical equipment of a computer system.

Categories of Hardware

Category	Examples
Input	Keyboard, mouse, scanner
Output	Monitor, printer, speaker
Processing	CPU, GPU
Memory	RAM, ROM, cache
Storage	Hard disk drive, solid-state drive
Communication	Network interface card, motherboard

Every physical part of the machine belongs to one of these groups.

What Is Software?

- ▶ **Software** is a set of **programs and instructions** that tell the hardware what to do.
- ▶ It is **intangible**: you cannot touch it.
- ▶ It is stored on secondary storage and loaded into memory when it runs.
- ▶ Two broad kinds: **system software** and **application software**.

Definition

Software is the collection of programs, procedures, and related data that direct the operation of a computer.

Hardware vs Software

Point	Hardware	Software
Nature	Physical, tangible	Logical, intangible
Touched?	Yes	No
Role	Executes work	Gives instructions
Example	CPU, printer	MS Word, driver
Failure	Wears out, breaks	Bugs, corruption

Both are needed: one without the other cannot run a computer.

System Software vs Application Software

- ▶ **System software** runs and manages the computer itself: the operating system, device drivers, and utilities.
- ▶ **Application software** helps the user do a task: MS Word, a browser, a game.
- ▶ The **operating system** is **system software**. (PYQ-06)

Trap alert

The operating system is system software, not application software.

What Is Firmware?

- ▶ **Firmware** is **software embedded in hardware**, stored in **non-volatile** memory.
- ▶ It is kept in ROM, EPROM, or flash memory inside the device.
- ▶ It gives a device its **low-level control** instructions and starts it up.
- ▶ Examples: **BIOS/UEFI** on a PC, and the firmware inside a printer, router, or keyboard.

Definition

Firmware is software permanently stored in a hardware device; it is software by nature, hardware by storage.

Firmware vs Ordinary Software

Point	Firmware	Software
Tied to	A specific device	General-purpose
Stored in	ROM / flash on the device	Secondary storage
Updated	Rarely, by flashing	Often, by install
Example	BIOS, printer firmware	MS Word, a driver

Firmware is software locked to one device; ordinary software is portable.

Device Driver

- ▶ A **device driver** is a small **program** that lets the operating system communicate with a piece of hardware.
- ▶ It **translates** general OS commands into the signals a specific device understands.
- ▶ It is **software**, although it is tied to a hardware device. (TRAP-10)
- ▶ Example: a **printer driver** lets the OS print on a particular printer model.

Trap alert

A device driver is **software**, not hardware. (TRAP-10)

Humanware

- ▶ **Humanware** (also called **peopleware**) means the **people** who use and run the computer system.
- ▶ It includes end users, programmers, system administrators, and operators.
- ▶ Hardware and software are useless without the human element.

Definition

Humanware is the human component of a computing system.

The Layered Computing Stack

Layer	Role
Hardware	The physical components
Firmware	Stored on the hardware; starts and controls it
Device driver	Lets the OS talk to the hardware
Operating system	Manages all resources and services
Application software	Lets the user do a task

Each layer relies on the layer below it.

Reading the Stack

- ▶ The stack runs from **hardware** at the bottom up to **application software** at the top.
- ▶ Order: Hardware → Firmware → Device driver → Operating system → Application.
- ▶ The **operating system** sits above the driver and firmware, and below the applications.
- ▶ A command from an application passes **down** the stack until it reaches the hardware.

Worked Example: Classify Each Item

- ▶ CPU → **hardware**.
- ▶ Windows (an operating system) → **system software**.
- ▶ A printer driver → **software**.
- ▶ BIOS chip code → **firmware**.
- ▶ MS Word → **application software**.
- ▶ The person using the PC → **humanware**.

Exam habit

Ask first: is it physical? If not, is it a program tied to a device (firmware), a helper for hardware (driver), or a user task (application)?

Trap Alert: Hardware or Software?

- ▶ A device driver is **software**, not hardware. (TRAP-10)
- ▶ The operating system is **system software**, not application software. (PYQ-06)
- ▶ Firmware is **software** stored in non-volatile memory inside hardware.
- ▶ Humanware means **people**, not a physical device.

Remember

Tangible = hardware. Instructions and programs = software.

Lesson 17: Recall

- ▶ Hardware is the physical equipment; software is the set of programs and instructions.
- ▶ System software manages the machine; application software does user tasks. The operating system is system software.
- ▶ Firmware is software embedded in hardware and stored in non-volatile memory, such as the BIOS.
- ▶ A device driver is software that lets the OS communicate with hardware.
- ▶ Humanware (peopleware) is the human component of the system.
- ▶ The stack, bottom to top: Hardware → Firmware → Device driver → Operating system → Application.